

STAT 1210-2

Introduction to Statistics for the Life and Social Sciences

Chapter 0: Course Information

Ali Raisolsadat

School of Mathematical and Computational Sciences
University of Prince Edward Island

Fall 2026

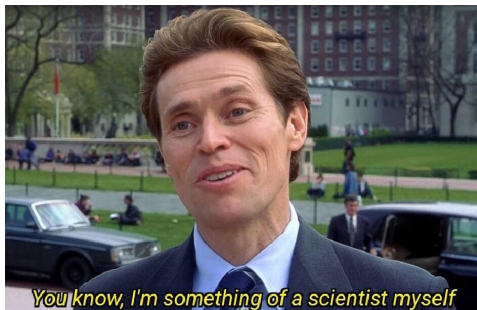
Introduction to Statistics for the Life and Social Sciences — Fall 2026

Instructor: Ali Raisolsadat
Office hours: TBD
E-mail: sraisolsadat@upei.ca
Lectures: T, Th 1:00 PM - 2:15 PM
Location: Irving Chemistry Centre, Room 104

Your Instructor

I am Ali Raisolsadat.

- Master of Mathematics (MMath) in Computational Mathematics, University of Waterloo
- Bachelor of Science in Actuarial Science, University of Prince Edward Island
- Bachelor of Science in Financial Mathematics, University of Prince Edward Island



Course Materials and Communication

Textbook

The Practice of Statistics in the Life Sciences by Brigitte Baldi and David S. Moore

Course website and Moodle

- The course website will contain the outline, lecture slides and supporting materials.
- Moodle will be used for official course communication, announcements and schedule updates and grades.
- Students are responsible for checking course announcements regularly.

Course Goal

Develop the statistical reasoning needed to organize data, describe patterns, quantify uncertainty and draw appropriate conclusions in life-science and social-science settings.

Course Roadmap: Exploring Data and Probability

In the first part of the course, we begin by learning how to describe data and using probability to model and understand uncertainty.

I. Exploring Data

- **Chapters 1–2:** Distributions and numerical summaries
- **Chapter 3:** Scatterplots and correlation
- **Chapter 4:** Regression

II. Probability and Distributions

- **Chapters 9–10:** Probability rules
- **Chapter 11:** Normal distributions
- **Chapter 12:** Binomial and Poisson distributions

Course Roadmap: Statistical Inference

In the second half of the course we use sample data to make conclusions about populations.

III. Foundations of Inference

- **Chapter 6:** Samples and observational studies
- **Chapter 13:** Sampling distributions
- **Chapter 14:** Confidence intervals and p -values
- **Chapter 15:** Inference in practice

IV. Specific Procedures

- **Chapter 17:** Inference about a population mean
- **Chapter 19:** Inference about a population proportion

How Your Final Grade Is Determined

Assessment	Weight
Four assignments (5% each)	20%
Examinations: midterm or quiz option	40%
Final examination	40%
Total	100%

Option A: Two Midterms

$$20\% + 20\% = 40\%$$

Two 75 minutes midterms.

Option B: Four Quizzes

$$10\% + 10\% + 10\% + 10\% = 40\%$$

Four 40 minutes quizzes.

Academic Integrity and the Use of AI Tools

ChatGPT, Claude, Gemini and other large language models may be used as study aids unless an assessment states otherwise. Students must not submit AI-generated answers as their own work or use these tools in ways that violate assessment instructions.

If your submitted work raises concerns about unauthorized assistance or academic dishonesty, I may ask you to explain the calculations, reasoning, interpretations and conclusions. You must be able to explain and defend the work they submit.

Failure to demonstrate understanding may result in a substantial mark reduction and may be addressed under the University's academic integrity procedures.



Optional 2% Course Introduction Task

Your Task

Write an approximately **250-word handwritten paragraph** introducing yourself and your goals for this course.

In your paragraph, address the following questions:

- What is your major and why did you choose it? What interests you about it?
- If you have not selected a major, what options are you currently considering? If you are beyond first year, explain why you chose your current major.
- How do you think STAT 1210 will help you and what do you want from the course? For example, are you interested in learning statistics, taking it mainly as a program requirement, or both?
- What final grade do you realistically think you will achieve in this course?

Optional 2% Course Introduction Task



Credit

A thoughtful response that follows the instructions will add **2% to your course grade**. Ensure to include your name and student ID.